Kernel-Level Memory Compression and System Performance Using ZRAM

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64.6%	0.21s	75%	40
CPU reduction at 75% pressure	Max speed difference	I/O wait reduction	Total test runs

RESEARCH QUESTIONS

RQ1 Does kernel-level memory compression improve performance under heavy workloads?

RQ2 What is the impact on CPU utilization when compression is used?

METHODOLOGY

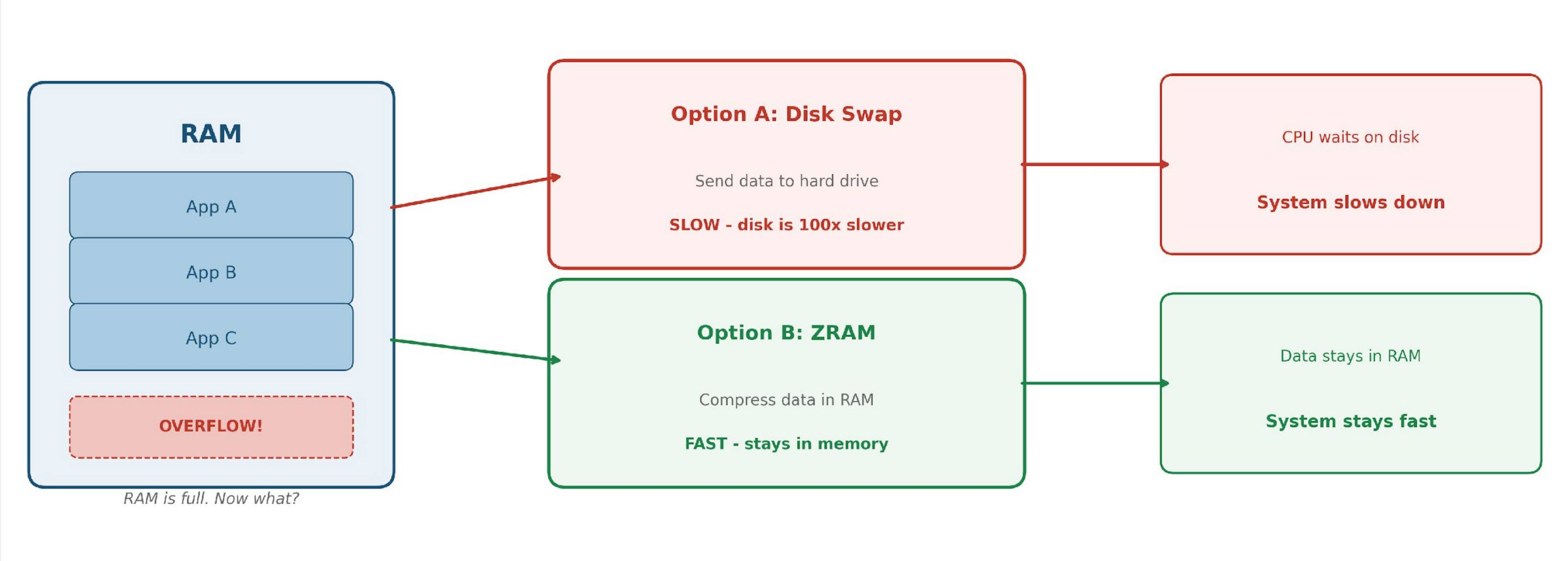
Test Environment

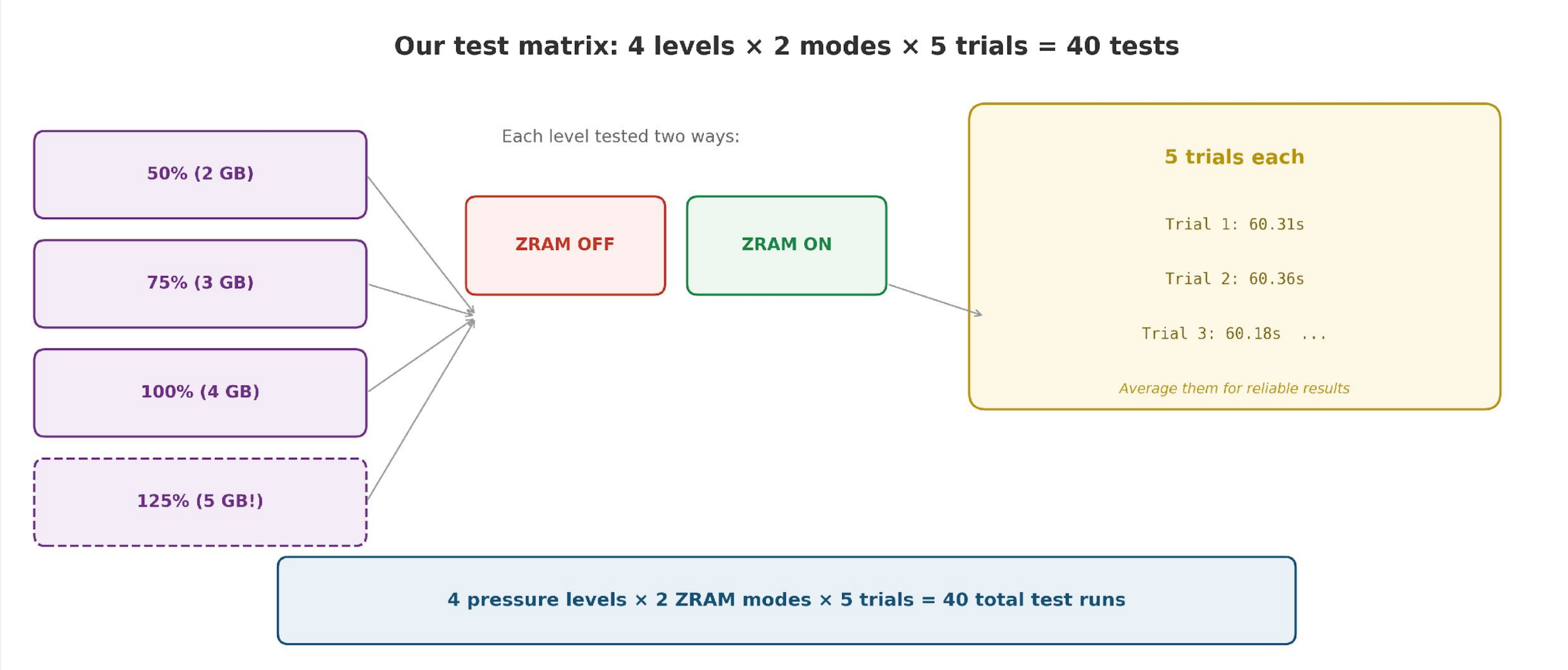
ThinkPad • 32 GB RAM • SSD • Fedora Linux
cgroups v2 limits memory to 4 GB
ZRAM with lz4 algorithm • stress-ng 2 workers

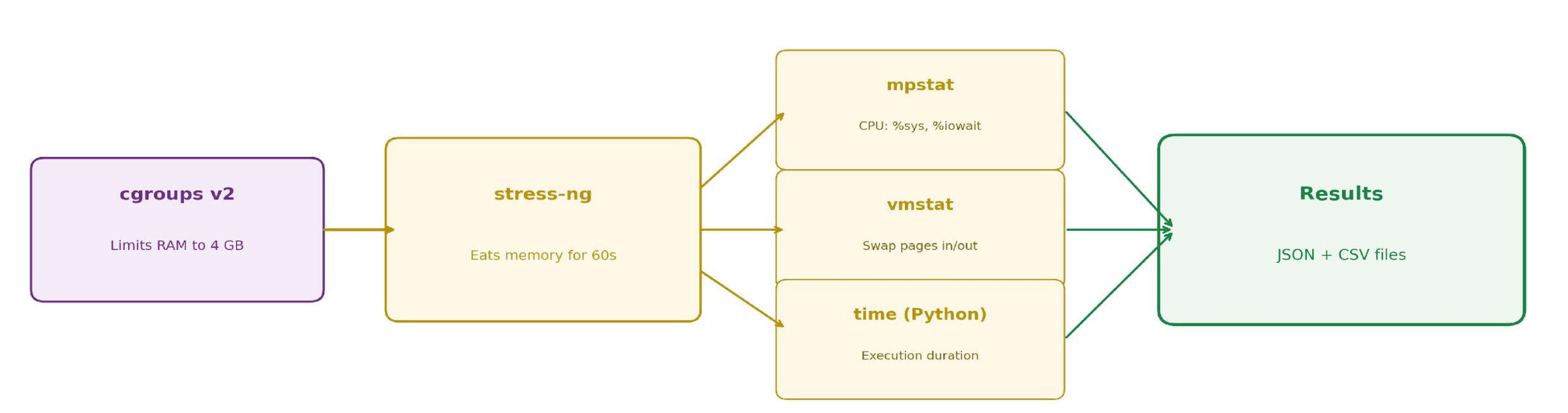
Two Phases

Phase 1: Measure execution time (throughput)
Phase 2: Measure CPU overhead (%sys, %iowait)

4 levels × 2 modes × 5 trials = 40 runs







ANALYSIS — RQ1: PERFORMANCE

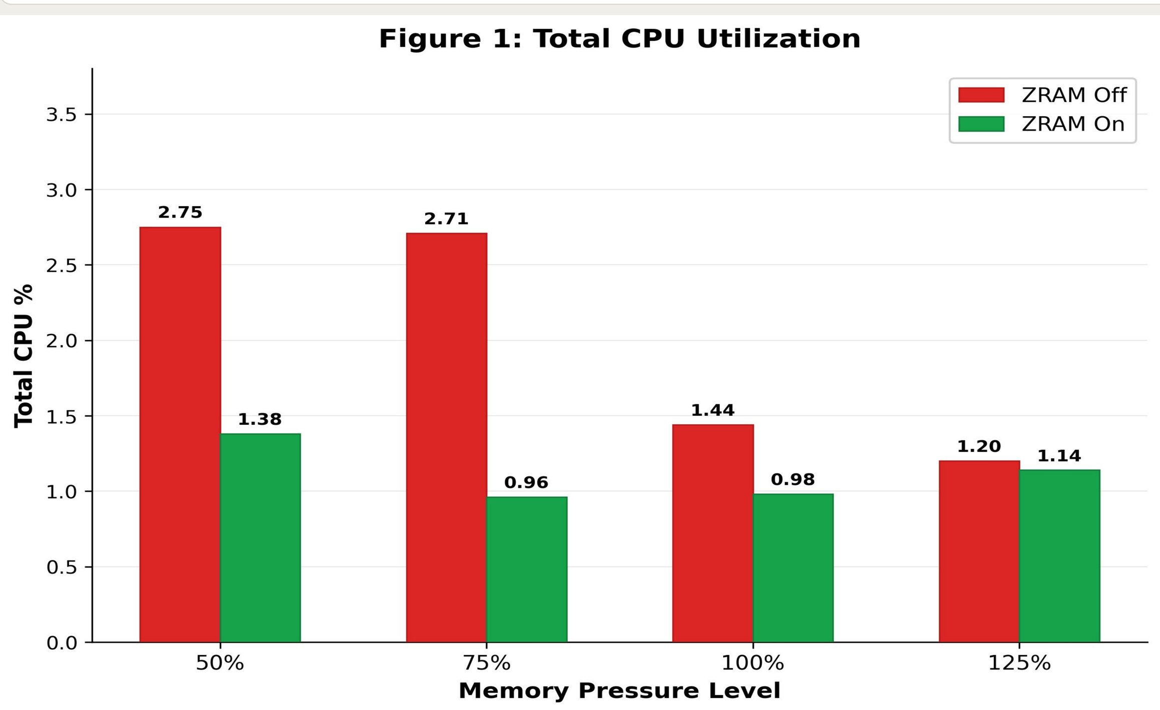
Execution times were virtually identical across all configurations

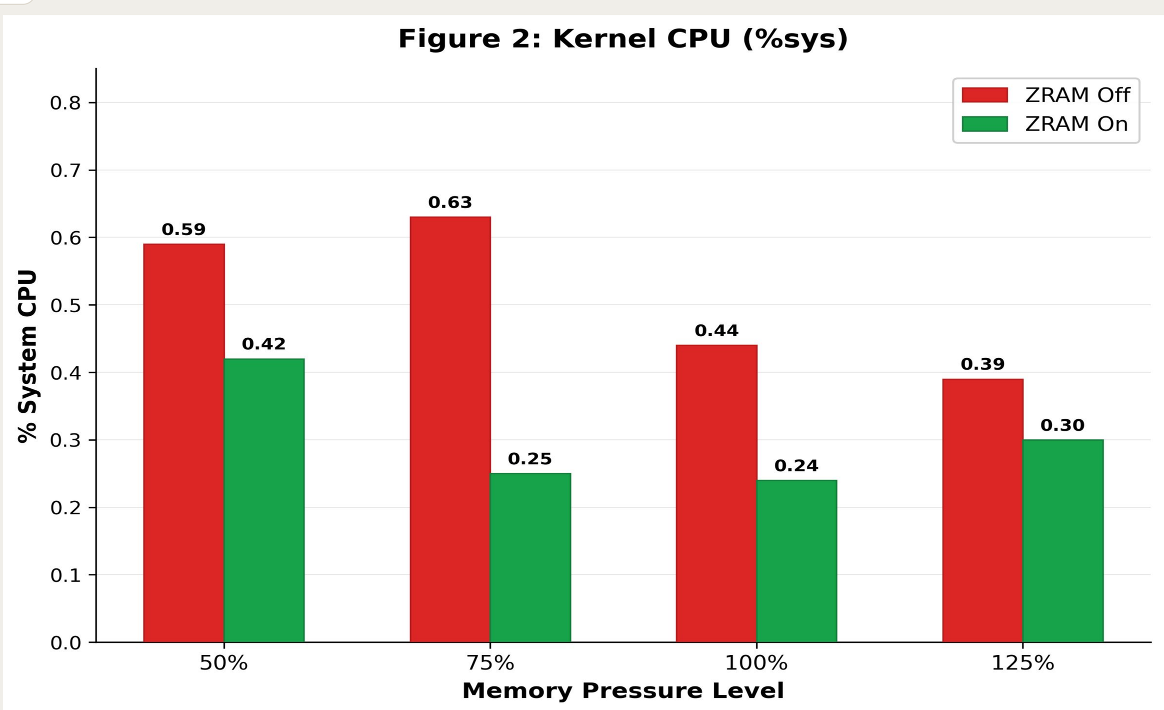
Range: 60.16s to 60.37s (only 0.21s difference)
ZRAM introduces zero performance penalty — safe to leave on

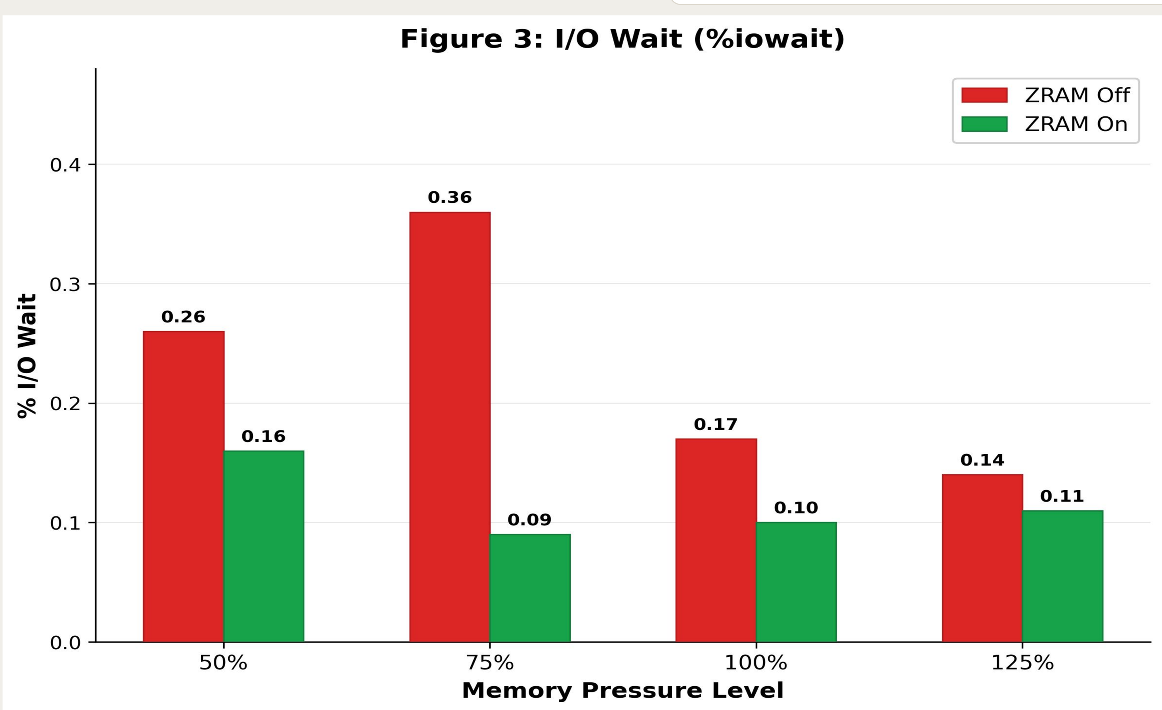
ANALYSIS — RQ2: CPU IMPACT

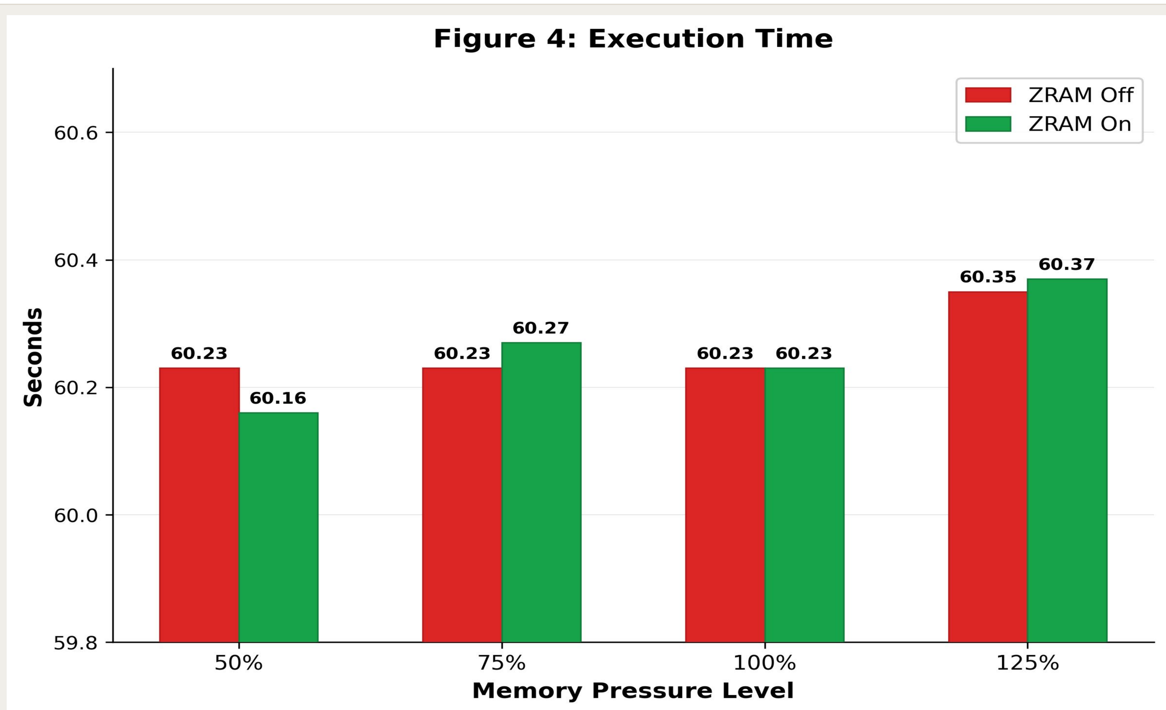
ZRAM reduced CPU usage at every pressure level

At 75% pressure: CPU dropped from 2.71% → 0.96% (−64.6%)
Kernel CPU (%sys) consistently lower with ZRAM
I/O wait dropped up to 75% — less time waiting on disk
lz4 compression is cheaper than disk swap management









Red = ZRAM Off • Green = ZRAM On • All figures show results across four memory pressure levels (50%, 75%, 100%, 125%)

CONCLUSION

1
ZRAM does not degrade system throughput
Execution times were identical regardless of ZRAM state

2
CPU reduced 5–65% at every pressure level
Strongest benefit at moderate pressure (75%)

3
lz4 is cheaper than disk swap management
Lower kernel CPU and lower I/O wait

4
ZRAM is a safe default-on configuration
CPU benefits with no penalty under any tested condition